Peter Dietrich
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DTE

10 CFR 50.73

November 22, 2019
NRC-19-0077

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, DC 20555-0001

Fermi 2 Power Plant
NRC Docket No. 50-341
NRC License No. NPF-43

Subject: Licensee Event Report (LER) No. 2019-005

Pursuant to 10 CFR 50.73(a)(2)(v)(C), DTE Electric Company (DTE) is submitting LER No. 2019-005, Secondary Containment Pressure Exceeded Technical Specification Due to Reactor Building HVAC Damper Malfunction.

No new commitments are being made in this submittal.

Should you have any questions or require additional information, please contact Mr. Jason R. Haas, Manager – Nuclear Licensing, at (734) 586-1769.

Sincerely,

Peter Dietrich
Senior Vice President and CNO

Enclosure: Licensee Event Report No. 2019-005, Secondary Containment Pressure Exceeded Technical Specification Due to Reactor Building HVAC Damper Malfunction

cc: NRC Project Manager
NRC Resident Office
Regional Administrator, Region III
Michigan Department of Environment, Great Lakes, and Energy

Enclosure to
NRC-19-0077

Fermi 2 NRC Docket No. 50-341
Operating License No. NPF-43

Licensee Event Report (LER) No. 2019-005, Secondary Containment Pressure Exceeded Technical Specification Due to Reactor Building HVAC Damper Malfunction

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
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4. Title

Secondary Containment Pressure Exceeded Technical Specification Due to Reactor Building HVAC Damper Malfunction

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
									N/A	05000
09	29	2019	2019	- 005	- 00	11	22	2019	Facility Name N/A	Docket Number 05000

9. Operating Mode: 1

10. Power Level: 100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☑ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
Fermi 2 / Jason R. Haas – Manager, Nuclear Licensing	(734) 586-1769

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	VA	62	Agastat	Y	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected	15. Expected Submission Date	Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date) ☑ No				

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On September 29, 2019, at 2228 EDT, a planned train swap of the Reactor Building Heating Ventilation and Air Conditioning (RBHVAC) system resulted in the Technical Specification (TS) for secondary containment (SC) pressure boundary not being met for approximately 2 minutes and 15 seconds. The maximum secondary containment pressure recorded during that time was approximately 0.1 inches of water gauge (positive). Secondary containment pressure was restored to within TS limits by restarting the West RBHVAC train. There were no safety consequences or radiological releases associated with this event. The cause of this momentary loss of SC was determined to be the effect of the RBHVAC West exhaust fan discharge damper failing to close due to sticking of a time delay relay. For corrective actions, Fermi 2 will replace the relay and has temporarily modified the RBHVAC swap sequence until the relay has been replaced.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Fermi 2	05000- 341	2019	- 005	- 00

NARRATIVE

INITIAL PLANT CONDITIONS

Mode – 1
Reactor Power – 100%

There were no structures, systems, or components (SSCs) that were inoperable at the start of this event that contributed to this event.

DESCRIPTION OF THE EVENT

On September 29, 2019, while in Mode 1 at 100 percent power, a planned train swap of the Reactor Building Heating Ventilation and Air Conditioning (RBHVAC) [[VA]] trains was initiated to shut down the West train and place the East train in service. Following shutdown of the West RBHVAC train, the West exhaust fan [[FAN]] discharge damper [[DMP]] failed to close, resulting in an increase in secondary containment (SC) [[NH]] pressure. At 2228 EDT, the Technical Specification (TS) for SC pressure boundary was exceeded. The West RBHVAC train was restarted and restored SC vacuum to within the TS operability limit of greater than or equal to 0.125 inches of vacuum water gauge per TS Surveillance Requirement (SR) 3.6.4.1.1. It was noted that upon restart of the West RBHVAC train, the West exhaust fan discharge damper closed and then reopened normally during the train startup. The duration of the TS not being met was approximately 2 minutes and 15 seconds and the maximum SC pressure observed during the event was approximately 0.1 inches of water gauge (positive). SC was declared OPERABLE at 2235 EDT.

An 8-hour event notification (EN 54299) was made to the NRC based on meeting the reporting criteria of Title 10 Code of Federal Regulations (10 CFR) 50.72(b)(3)(v)(C) as an event or condition that could have prevented the fulfillment of a safety function needed to control the release of radioactive material. This Licensee Event Report (LER) is being made under the corresponding requirement in 10 CFR 50.73(a)(2)(v)(C).

SIGNIFICANT SAFETY CONSEQUENCES AND IMPLICATIONS

There were no safety consequences or radiological releases associated with this event. At no time during this event was there a potential for endangering the public health and safety.

The specified safety function of the SC is to contain, dilute, and hold up fission products that may leak from primary containment following a Design Basis Accident (DBA). In conjunction with operation of the Standby Gas Treatment System (SGTS) [[BH]] and closure of certain valves [[V]] whose lines penetrate the SC, the SC is designed to reduce the activity level of the fission products prior to release to the environment and to isolate and contain fission products that are released during certain operations that take place inside primary containment, when primary containment is not required to be OPERABLE, or that take place outside primary containment. It is possible for the pressure in the control volume to rise relative to the environmental pressure (e.g., due to pump [[P]] and motor [[MO]] heat load additions). To prevent ground level exfiltration while allowing the SC to be designed as a conventional structure, the SC requires support systems to maintain the control volume pressure at less than the external pressure. For the SC to be considered OPERABLE, it must have adequate leak tightness to ensure that the required vacuum can be established and maintained.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
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NARRATIVE

During this event, a higher indicated SC pressure was recorded for approximately 2 minutes and 15 seconds. In Chapter 15 of the Updated Final Safety Analysis Report (UFSAR), RBHVAC is assumed lost at the onset of a Loss of Coolant Accident (LOCA) concurrent with a Loss of Offsite Power. As a result, calculations show that the SC would be pressurized until the SGTS restores vacuum. The structural integrity (i.e., leak tightness) of the SC was re-confirmed on September 29 when SC vacuum was restored to greater than 0.125 inches of vacuum water gauge by restarting an RBHVAC train.

If the DBA LOCA concurrent with a Loss of Offsite Power had occurred during the time when the SC pressure TS limit was exceeded, the SC was sufficiently leak tight such that the SGTS would still have established and maintained vacuum greater than the TS required value without RBHVAC in service.

The radiological consequences of the DBA LOCA for SC contained in Chapter 15 of the Fermi 2 UFSAR result in doses that are below 10 CFR 50.67. The SC is assumed to be at a pressure of 0.0 inches of vacuum water gauge at the onset of the LOCA. For this particular event, had the DBA LOCA for SC actually occurred, the increase in magnitude of radiological dose as a result of increased draw-down time from starting at 0.10 inches of water gauge instead of 0.0 inches of water gauge, would be minimal and negated by several conservative assumptions in the existing analysis (e.g., 100% exfiltration from SC during the first 15 minutes of drawdown with SGTS in operation, 10% exfiltration from SC with SGTS in operation throughout the remaining 30 day duration of the accident, no holdup time in SC throughout the 30 day duration of the accident, and all exfiltration and filtered releases are at ground level). These conservative assumptions are not reflective of actual plant conditions and configurations.

CAUSE OF THE EVENT

The most probable cause of the event was intermittent sticking of a time delay relay [[62]] (Agastat timer relay model 7012AD) associated with the West exhaust fan discharge damper due to the relay approaching its end of life. The relay sticking prevented the West exhaust fan discharge damper from closing as required during West RBHVAC train shutdown. This resulted in increased pressure within the Reactor Building.

CORRECTIVE ACTIONS

Immediate corrective actions included restarting the West RBHVAC train to restore SC vacuum to within the TS operability limit. When the West RBHVAC train was restarted, the West exhaust fan discharge damper cycled normally.

The time delay relay is scheduled for replacement in accordance with the Fermi 2 corrective action program. Until the relay is replaced, the RBHVAC train swap sequence has been temporarily modified to eliminate the potential that the relay sticking could cause SC pressure transients to exceed TS limits. In addition, the preventive maintenance (PM) frequency of the time delay relays will be increased to ensure the relays are replaced with sufficient margin such that intermittent sticking does not occur as they approach their end of life.

NARRATIVE

PREVIOUS OCCURRENCES

Events involving loss of SC due to the issues with the RBHVAC system have been reported in past, including the following recent LERs:

LER 2015-001 involved the loss of SC function due to an RBHVAC system trip caused by a valid actuation of a freeze protection device.

LER 2015-004 involved the loss of SC function due to reverse rotation of the RBHVAC center exhaust fan during post-maintenance testing caused by reversed electrical leads.

LER 2015-005-01 involved the loss of SC function due to setpoint drift of the RBHVAC supply damper time delay relay resulting in the dampers moving out of sequence.

LER 2016-005 involved the loss of SC function due to the combined effect of high winds during the RBHVAC startup sequence.

LER 2018-001 involved the loss of SC function due to an RBHVAC exhaust fan modulating damper failing to full open.

Of the LERs listed above, LER 2015-005-01 and LER 2016-005 involved issues with the time delay relays. Previous corrective actions were taken to address relay setpoint drift and timing. The reason the corrective actions taken in response to these LERs did not prevent the current event from occurring is that the intermittent sticking of the relay in this LER was related to the relay approaching its end of life and this particular relay did not need to be replaced in response to any of these previous LERs.